

PAPER_1682_DM_DIRECT_FLOOR_LAMBDA4

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PAPER_1682 — DM Direct-Detection Floor = $\sigma_{\text{floor}} = \Lambda \times 10 \text{ cm}^2$ (predicts null detections)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Dark Matter

Date: June 18, 2026

Status: CLOSED — EXACT formula closure (PAPER_145x-147x broader corpus)

Observation

PAPER_1454 (PAPER_14xx broader corpus, classic cosmology/BSM puzzle) gives:

``

$\sigma_{\text{floor}} = \Lambda \times 10 \text{ cm}^2$ (predicts null detections)

``

Status: **EXACT formula.**

UQFF Closed Identity

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$\sigma_{\text{floor}} = \Lambda \times 10 \text{ cm}^2$ (predicts null detections) EXACT formula

``

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Empirical dark matter measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1454

- Related: PAPER_1666-1675 (tier-25); PAPER_1656-1665 (tier-24)

- Calculator dispatch: `calculate_paradox({"paradox": "dm_direct_floor_lambda4"})`

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